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COMPARISONS OF OBSERVATIONAL ANGLES BETWEEN MOON-BASED PLATFORM AND ARTIFICIAL SATELLITES

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ABSTRACT

The Moon-based platform has different observation angular characteristics from artificial satellites due to its peculiar orbit. To illustrate the difference, we first normalize the positions and attitudes of the Sun, Earth, and Moon into the same reference system using coordinate transformations. Then, we calculate the observation angles (including viewing elevation angel, solar elevation angel and relative azimuth angle) distribution of satellites in LEO, GEO and at Lagrange L1 satellites during the same period. Furthermore, the observation angles distribution of the Moon-based platform in different seasons with the Earth observed point located at different positions are analyzed. Results revealed that compared to artificial satellites, the Moon-based platform has continuous viewing observation angles and solar observation angles. Its observation angles change with the seasons, and it provides more angular information for equatorial regions than polar regions of the Earth.

Index Terms — Moon-based Earth observation, observation angular characteristics, observation angular analysis model

1. INTRODUCTION

In recent years, Earth observation technologies are increasingly attracting attention. Since the 1960s, a variety of artificial satellites have been designed and launched successfully. However, with the growing demand of Earth system science research, the existing artificial satellites are not yet capable of solving large-scale geoscientific problems well. Nowadays, a Moon-based platform is proposed to realize long-term and large-scale observation[1]. Furthermore, quantitative remote sensing requires multi-angle information for data. The Moon-based platform has different lunar orbit from traditional satellites, which lead to its peculiar observation angular characteristics.

The orbit of Moon-based platform has the following characteristics [2]. First, the average distance between the Moon and the Earth is 380,000 km which allows the Moon-based platform to provide unique observation of the Earth Moon-facing hemisphere; Second, for the Moon-based platform, the orbital altitude isn't consistent; Third, the value of the angle between the orbital plane of the Moon and the equatorial plane of the Earth is not consistent. Therefore, compared with traditional satellites, the Moon-based platform has the special orbit characteristics of almost hemisphere observation, variable orbital altitude and inclination, which determine that Moon-based platform has different observation angular characteristics.

The observation angle is one of the most important parameters of Moon-based platform geometry. The typical observation angle parameters mainly include the viewing elevation angle, the solar elevation angle and the relative azimuth angle [3]. The elevation angle means the angle between a direction of interest and the local horizon, and the azimuth angle is the angle between the north vector and the vector of the object of interest on the horizontal plane. Some pioneer research work on observation angels of the Moon-based platform has been done. Guo *et al.* [4] developed an angular geometric model of Moon-based Earth observations based on four observation angles (viewing and solar zenith angles, viewing and solar azimuth angles). Shen *et al.* [5] studied the changes of local azimuth-elevation angle of Earth observed point at different times in 2016. Ye *et al.* [6] proposed a new angular geometric model which reduced the solar azimuth angle and viewing azimuth angle to the relative azimuth angle. As a key feature of observation geometry, the analysis of the observation angles helps expand potential application of the Moon-based platform [7].

In this study, we focus on the angular characteristics of the Moon-based platform and other artificial satellites. The differences of them are discussed. Further, the effect of different season and location of Earth observed point is analyzed using the observation angular analysis model.

2. METHODOLOGY

As mentioned above, the Moon-based platform has unique angular characteristics. To illustrate the difference, the observation angular analysis model is established. First, the Moon-based platform geometrical model is developed. Then, the theoretical expressions of observation angles including

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viewing elevation angle, solar elevation angle and relative azimuth angle are deduced. Fig.1 summarizes the general procedures of the observation angular analysis model.

2.1. Observation geometrical model

The transformation between coordinate systems involved in the Moon-based platform is established aiming to transform the positions of Earth, Moon and Sun into a unified reference system [8]. The coordinate systems can be divided into three categories. The location of the lunar platform is defined in the lunar reference system, the location of the object of interest on Earth is defined in the Earth reference system, and in the celestial reference system, the relative positions of the Earth, Moon and Sun are defined. The transformation from the Lunar topocentric reference system (C_{LTRS}) to ITRS (C_{ITRS}) can be calculated as:

$$C_{ITRS} = [\Pi][\Omega][T][M][\Gamma][B][\Theta][R]C_{LTRS} \quad (1)$$

Where $[\Pi], [\Omega], [T], [M], [\Gamma], [B], [\Theta]$, and $[R]$ represent polar motion, rotation, nutation, precession, bias matrix, offset of the origin from the SCRS to the GCRS, lunar libration, constant, the rotation matrix from the LTRS to SCRS.

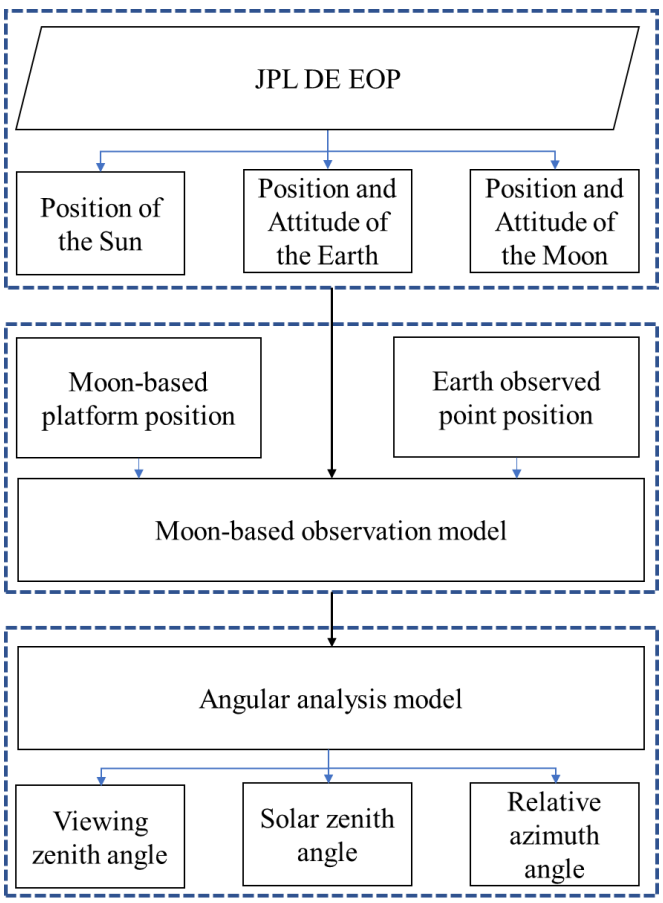

Fig.1. The scheme of the observation angular analysis model

2.2. Observation angular analysis model

The observation angles including viewing elevation angle, solar elevation angle and relative azimuth angle are defined and calculated in the LTRS [9].

1. Viewing elevation angle. It is the angle between the line of sight from the sensor of the Moon-based platform (L_m) to the observed point on Earth and local horizon. The viewing elevation angle (h_v) of Moon-based platform can be calculated as:
$$h_v = \frac{\pi}{2} - arccos(\frac{L_m \cdot n}{|L_m||n|}) \quad (2)$$
where n is the normal vector of the local horizon plane.
2. Solar elevation angle. It is the angle between the solar direction (L_s) and local horizon. The solar elevation angle (h_s) can be calculated as:
$$h_s = \frac{\pi}{2} - arccos(\frac{L_s \cdot n}{|L_s||n|}) \quad (3)$$
3. Relative azimuth angle. It is the angle between the projection direction (φ_s) of the solar direction on the horizontal plane and the projection direction (φ_m) of the line of sight from the sensor of the Moon-based platform to the observed point on the horizontal plane, and it ranges from 0° to 180°. The relative azimuth angle (φ) can be calculated as:
$$\varphi = arccos(\frac{\varphi_s \cdot \varphi_m}{|\varphi_s||\varphi_m|}) \quad (4)$$

3. RESULTS

For comparison, we calculated the observation angles of satellites in LEO, in GEO and at L1 point in the Sun-Earth system during the same period. Further, we analyzed the observation angles of Moon-based platform in different seasons with the Earth observed point located at different positions.

The observation angles of satellites in LEO [see Fig.2(a)], GEO [see Fig.2(b)] and at L1 point [see Fig.2(c)] are calculated as shown in Fig.2. The observed point is (0°, 0°), and the observation period is from January 1, 2020 to January 30, 2020. It should be noted that for satellites in LEO, we have considered the limitation of field of view angle.

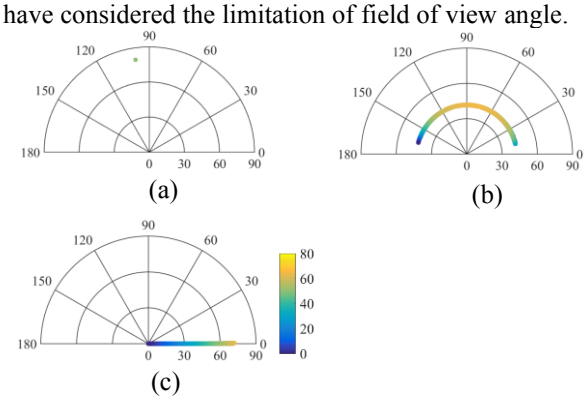

Fig.2. Observation angular distribution of satellites in (a) LEO, (b) GEO and (c) L1 point.

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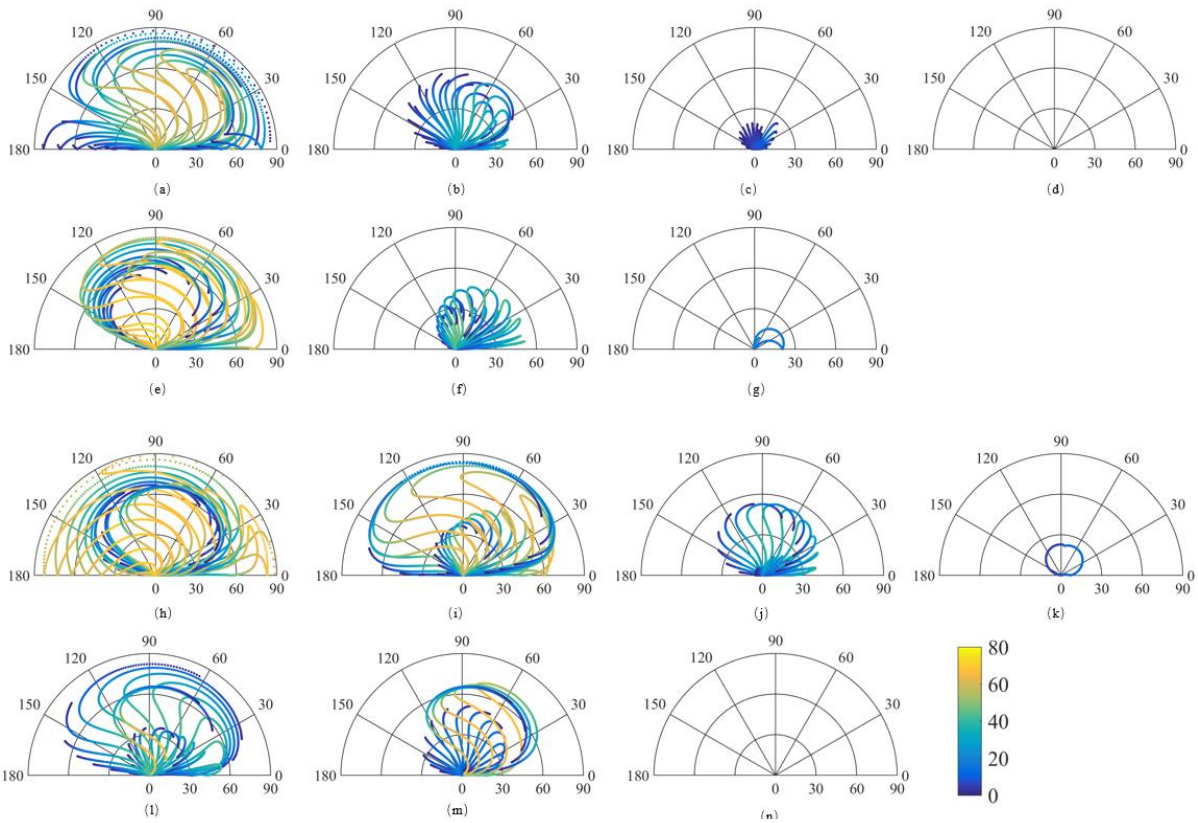

Fig.3. Observation angular distribution for Moon-based platform in January with the Earth observed point located at(a) (0°, 0°), (b) (0°, 30°N), (c) (0°, 60°N), (d) (0°, 90°N), (e) (0°, 30°S), (f) (0°, 60°S), (g) (0°, 90°S) and in April with the Earth observed point located at (h) (0°, 0°), (i) (0°, 30°N), (j) (0°, 60°N), (k) (0°, 90°N), (l) (0°, 30°S), (m) (0°, 60°S), (n) (0°, 90°S). (polar axis represents relative azimuth angles, polar diameter represents viewing elevation angles, and the color represents solar elevation angles).

As shown in Fig.2(a), for the satellites in LEO, this observed point can only be seen at 10:28 a.m. on January 3, when the viewing elevation angel is 80°, the solar elevation angle is 57°, and the relative azimuth angle is 98°. Fig.2(b) shows that for the satellites in GEO, the values of viewing elevation angels remain constant at 42°, and the values of solar elevation angles vary from 12° to 170°. Meanwhile, the values of relative azimuth angles vary from 0° to 72°. In Fig.2(c), for the satellite at L1 point, the values of solar elevation angles and the viewing elevation angels vary from 0° to 72°, and the relative azimuth angle is 0°.

Fig.3 and Fig.4 illustrate the observation angles of Moon-based platform with the Earth observed point located at (0°, 0°), (0°, 30°N), (0°, 60°N) , (0°,90°N) and (0°, 30°S), (0°, 60°S) , (0°, 90°S) in January, April, July and October, respectively. As can be inferred from the results, the observation angles of the Moon-based platform vary with the seasons, especially for the polar regions of the Earth. For example, the Moon-based platform cannot observe the Earth observed point of (0°, 90°N) in January and October, and it cannot observe the Earth observed point of (0°,90°S) in April and July. Furthermore, Moon-based platform provides more angular information for equatorial regions than high-latitude regions of the Earth. Compared to artificial satellites, the observation angles of Moon-based platform are continuous and more abundant.

To sum up, the different Earth observation platforms have peculiar observation angle characteristics due to their orbital characteristics. In LEO, satellites in Sun-synchronous orbit have constant Sun observation angles and viewing observation angles (considering the limitation of field of view angle). For the satellites in GEO, they have continuous solar observation angles and constant viewing observation angles. In particular, the satellite at the Lagrange L1 point has continuous viewing observation angles and constant relative azimuth angle. More specifically, the Moon-based platform has continuous viewing observation angles and solar observation angles, which provides diversified angular sampling data.

4. CONCLUSION

In this paper, the characteristics of the observation angles of the Moon-based platform and artificial satellites including satellites in LEO, GEO and at Lagrange L1 point are revealed

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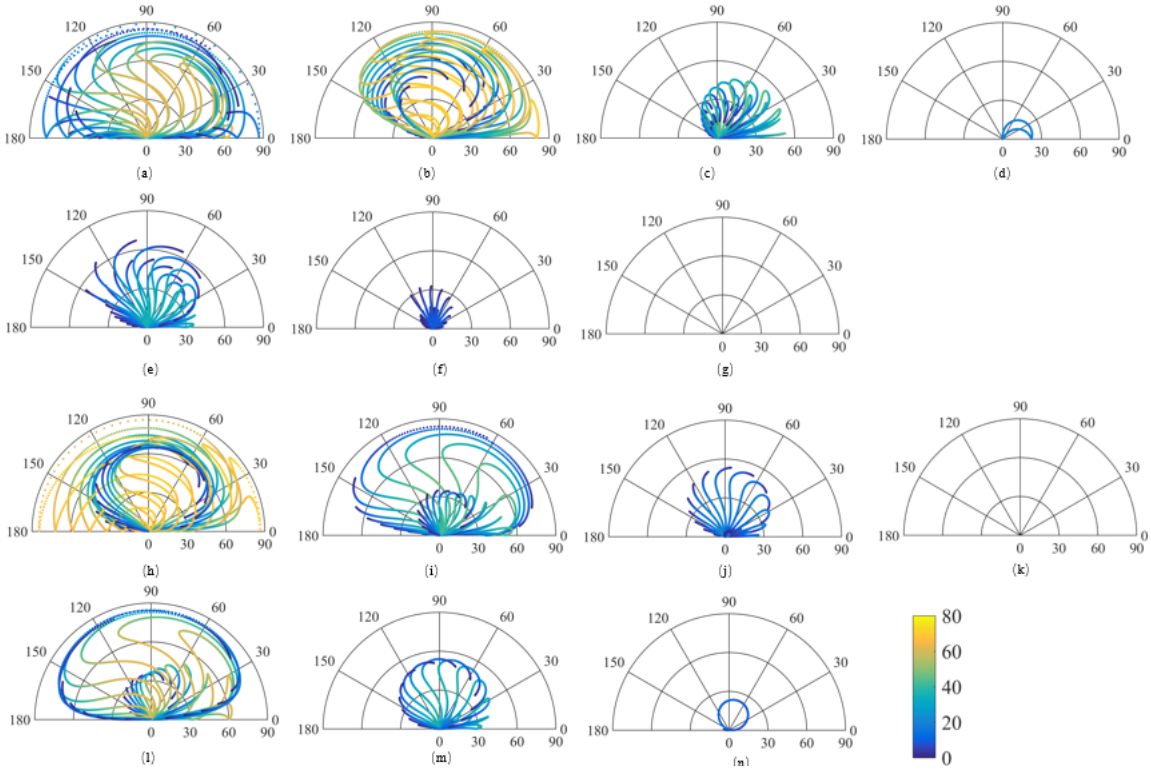

Fig.4. Observation angular distribution for Moon-based platform in July with the Earth observed point located at(a) (0°, 0°), (b) (0°, 30°N), (c) (0°, 60°N), (d) (0°, 90°N), (e) (0°, 30°S), (f) (0°, 60°S), (g) (0°, 90°S) and in October with the Earth observed point located at (h) (0°, 0°), (i) (0°, 30°N), (j) (0°, 60°N), (k) (0°, 90°N), (l) (0°, 30°S), (m) (0°, 60°S), (n) (0°, 90°S). (polar axis represents relative azimuth angles, polar diameter represents viewing elevation angles, and the color represents solar elevation angles).

and compared. By analyzing the sampling distribution of the three observation angles, we find that the observation angles information of Earth observation platforms with different orbit characteristics is discrepant, and the observation angles sampling of low-orbit satellites is limited by the field of view. Further, we compare the observation angles of the Moon-based platform in different seasons with the Earth observed point located at different positions. It is found that, the Moon-based platform has continuous viewing observation angles and solar observation angles. Its observation angles vary with the seasons, and it provides more angular information for equatorial regions than polar regions of the Earth. The analysis of the observation angles characteristics of the Moon-based platform is necessary for us to further expand the application field of the multi-angle data of the Moon-based platform.

5. REFERENCES

[1]H. Guo, G. Liu, and Y. Ding, "Moon-based Earth observation: scientific concept and potential applications," *International Journal of Digital Earth*,vol.11,no.6,pp.546-557,2017.

[2]H.Ye, H. Guo,G.Liu, and Y.Ren,"Observation scope and spatial coverage analysis for earth observation from a Moon-based platform," *International Journal of Remote Sensing*,vol.39,pp.5809-5833,2017.

[3] H.Ye, H. Guo, G.Liu, and Y.Ren,"Observation duration analysis for Earth surface features from a Moon-based platform,"*Advances in Space Research*,vol.62,no.2,pp.274-287,2018.

[4] H. Guo,Y.Ren, G.Liu and H.Ye,"The angular characteristics of Moon-based Earth observations," *International Journal of Digital Earth*,pp.339-354,2019.

[5]G.Shen, H.Guo, G.Liu,L.Zhang,and J.Huang,"Geometry numerical simulation and analysis for Moon-based Earth observation,"*IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*,vol.13,pp:3381-3393,2020.

[6]H.Ye,W.Zheng,H.Guo,G.Liu,"Effects of Temporal sampling interval on the Moon-based Earth observation geometry,"*IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*,vol.13,pp:4016-4029,2020.

[7]Y.Ren,H.Guo,G.Liu and H.Ye, "Simulation Study of Geometric Characteristics and Coverage for Moon-Based Earth Observation in the Electro-Optical Region," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol.10,no.6,pp.2431-2440,2017.

[8]H.Guo,G.Liu, and Y.Ding et al, "Moon-based Earth observation for large scale geoscience phenomena," *IEEE International Geoscience and Remote Sensing Symposium*,pp.3705-3707,2016.

[9] H.Ye, H. Guo, and G.Liu et al,"Observation angular analysis from a Moon-Based Earth observation platform,"*IEEE International Geoscience and Remote Sensing Symposium*, 14 November 2019.

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